

This is a medium-sized test document for performance benchmarking.

It contains multiple paragraphs discussing various aspects of Retrieval-Augmented Generation (RAG) systems.

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Section 2: Architecture

A typical RAG system consists of four main components:

1. Document ingestion pipeline (PDF parsing, chunking, embedding)
2. Vector storage system (for efficient similarity search)
3. Retrieval mechanism (to find relevant context)
4. Generation component (LLM that produces grounded answers)

Section 3: Embedding Models

Embedding models convert text into numerical vectors that capture semantic meaning. Common choices include:

- OpenAI's text-embedding-3-small (384 dimensions)
- HuggingFace's all-MiniLM-L6-v2 (384 dimensions)
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Section 4: Vector Stores

Vector stores enable efficient similarity search in high-dimensional spaces.

Popular options include:

- Qdrant (used in this project)
- FAISS
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Important security aspects of RAG systems:

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Emerging trends in RAG research:

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Vector stores enable efficient similarity search in high-dimensional spaces.

Popular options include:

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Embedding models convert text into numerical vectors that capture semantic meaning. Common choices include:

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Vector stores enable efficient similarity search in high-dimensional spaces.

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